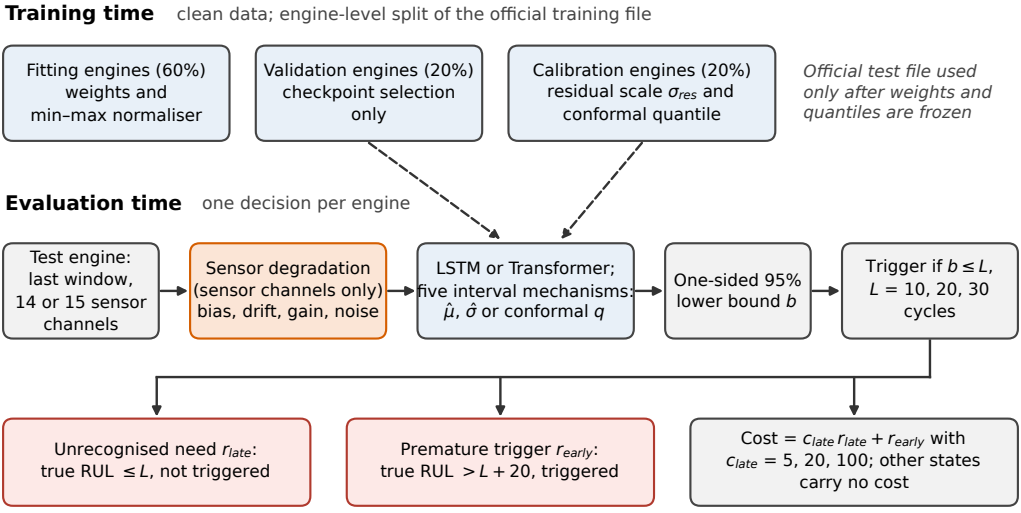

Training time
clean data; engine-level split of the official training file
Fitting engines (60%) weights and min–max normaliser
Validation engines (20%) checkpoint selection only
Calibration engines (20%) residual scale σ_{res} and conformal quantile
Official test file used only after weights and quantiles are frozen
Evaluation time
one decision per engine
Test engine: last window, 14 or 15 sensor channels
Sensor degradation (sensor channels only) bias, drift, gain, noise
LSTM or Transformer; five interval mechanisms: $\hat{\mu}$, $\hat{\sigma}$ or conformal q
One-sided 95% lower bound b
Trigger if $b \le L$, L = 10, 20, 30 cycles
Unrecognised need r_{late}: true RUL $\le L$, not triggered
Premature trigger r_{early}: true RUL $> L + 20$, triggered
Cost = $c_{late}\, r_{late} + r_{early}$ with c_{late} = 5, 20, 100; other states carry no cost